DriveCompose：可组合驾驶场景的生成式重建

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Abstract

在新的相机和交通参与者状态下复用驾驶场景，会暴露原始日志未曾记录的表面。重建这样的场景，需要恢复已观测内容、补全缺失的几何与外观，并为各个可移动组件保持稳定的身份。我们提出 *DriveCompose*，从有标定的图像序列中生成式地重建可组合驾驶场景。共享视觉编码器估计几何、资产身份和运动；我们将这些证据聚合到世界空间的背景查询及车辆、人体的规范查询，并编码为结构化潜表示。条件流模型在场景组件之间交换上下文，补全未知潜码，生成过程受已见潜码和已知自由空间约束。类型专属解码器输出高斯背景与资产，随后在输入视角下检查重建结果。学习过程依次包括视觉恢复、潜表示拟合、补全先验训练和基于渲染观测的适配。重建场景通过显式的相机、刚体和骨架状态进行装配。这一形式为生成过程提供了持久的三维输出：同一份补全内容可以沿不同轨迹渲染，无需重新采样几何或外观。

1. 引言

驾驶数据集沿采集轨迹记录场景 [3, 49]。驾驶仿真和场景编辑要求重建结果在原始配置改变后仍然可用。移动相机会露出被车辆遮挡的道路，转动车辆会暴露未见的侧面，改变行人姿态则会使此前被遮挡的身体表面进入视野。这些操作既需要原始图像之外的几何与外观，也需要一种表示，使内容始终归属于同一个场景和资产身份。

近期前馈方法直接从时序观测恢复驾驶场景。STORM 通过估计场景流聚合动态高斯，DGGT 从无位姿图像联合恢复相机参数与动态场景内容 [5, 71]。StreetForward 和 PointForward 进一步研究时序处理及重建证据的组织方式 [8, 77]。对于后续需要移动或驱动的资产，还需将不同时刻的观测汇入稳定的局部表示，并为新状态暴露的表面提供内容。

可组合表示解决了前一项需求。Street Gaussians 将车辆与背景分离，OmniRe 为刚体和关节对象建立规范模型，HUGSIM 将场景内容接入控制驱动的状态更新 [7, 68, 80]。生成模型则通过资产先验、高斯补全或生成视图处理缺失内容 [26, 69, 75, 83]。结合这些能力，需要生成器明确区分属于背景和各个资产的内容。生成的表面必须具有明确的归属、坐标系和运动规则，才能成为可编辑场景的一部分。

我们将驾驶场景重建建模为可组合场景的条件生成。表示的组织单位对应后续操作的对象：世界空间中的背景、局部坐标中的刚体车辆，以及带骨架绑定的规范人体表面。图像证据先在这些支撑上聚合，生成模型再根据已观测场景条件补全缺失潜码。每个补全表面都在所属组件的控制坐标系中解码。

DriveCompose 通过视觉恢复前端、结构化潜码和条件流模型实现这一形式（图 1）。前端估计几何、轨迹级身份和随时间变化的状态，规范聚合将不同视角、不同时刻对同一内容的观测汇集起来。在补全模型中，局部注意力沿各组件的支撑组织，共享记忆在背景与资产之间交换上下文。每个查询同时携带规范坐标及其在参考世界配置中的位置：前者标识持久内容，后者表达与场景的空间关系。类型专属解码器将完成的潜码转化为高斯组件。

生成还需要遵循场景恢复所依据的图像。我们在采样过程中保留可靠的已见潜码，并抑制已知自由空间。新增内容仍可能遮挡已观测表面，因此还需在原始视角下检查装配后的候选场景。构建完成后，背景与规范资产保持固定，后续渲染由相机位置、车辆变换和人体关节姿态决定。学习过程遵循这一分工：先拟合恢复模块和潜表示，再训练补全先验，最后保持表示固定，通过可微渲染适配生成模型。

本文的贡献包括：

- 提出生成式重建框架，将驾驶观测组织为可组合高斯场景，以世界空间背景和持久规范资产支持状态驱动复用。
- 建立结构化补全模型，将类别专属支撑和场景上下文与已见潜码约束、自由空间抑制及渲染观测检查结合。

2. 相关工作

辐射场表示。 NeRF 通过连续辐射场表示几何与外观 [33]，后续方法扩展到非受控照片和多尺度采样 [1, 31]。高斯泼溅提供用于渲染的显式基元 [22]；Mip-Splatting 处理混叠 [76]，高斯光线追踪和 3DGUT 扩展射线处理与相机模型 [34, 63]。D-NeRF 和可变形高斯引入随时间变化的形变 [40, 74]。这些表示为场景分解与资产控制提供了渲染基础。

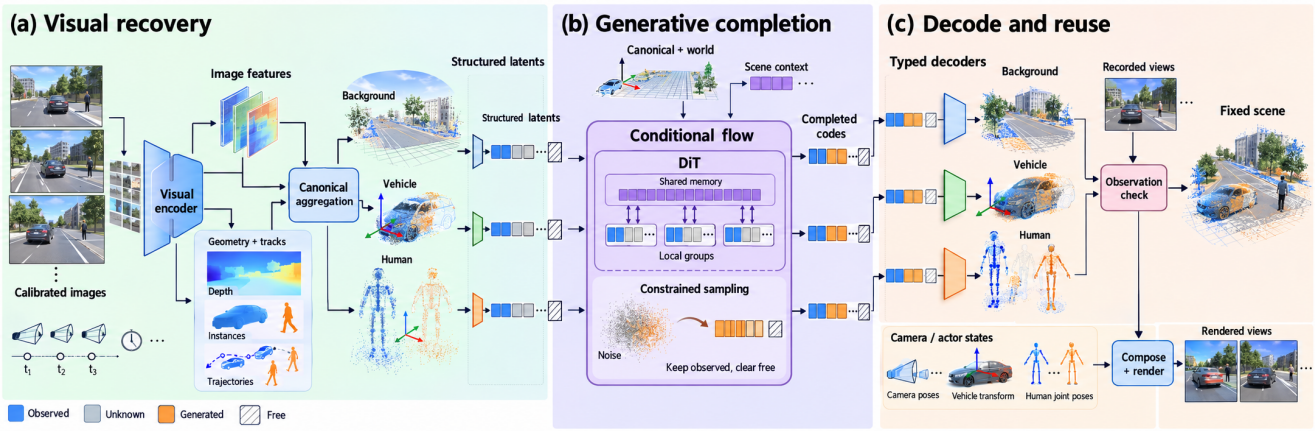


Figure 1. **DriveCompose 总览**。(a) 从有标定图像恢复几何、身份与运动证据，在背景和规范资产支撑上聚合并编码为结构化潜码。(b) 场景条件流 Transformer 补全未知潜码，同时保留已见内容并抑制自由空间。(c) 类型解码器输出高斯组件，在原始视角下检查后复用场景；相机与资产状态控制后续装配和渲染。图中场景为机制示意，不是实验输出。

城市场景与控制。 Urban Radiance Fields 和 Block-NeRF 面向大规模城市采集 [43, 50]。NeuRAD、EVol-Splat 和 SplatAD 将神经渲染适配到驾驶场景与传感器 [17, 32, 53]。Neural Scene Graphs 和 EmerNeRF 通过对象结构或自监督分解组织动态内容 [35, 70]。Street Gaussians、DrivingGaussian 和面向城市场景的 HUGS 区分前景与背景 [68, 81, 82]；OmniRe 将重建扩展到多类参与者 [7]。UniSim 和 HUGSIM 将重建场景接入闭环仿真 [73, 80]。我们在用于生成式补全的变量中保留组件身份与运动规则。

前馈重建。 LRM、LGM 和 GS-LRM 学习从图像到三维的预测 [18, 51, 78]。pixelSplat、MVSplat 和 DepthSplat 从稀疏视图恢复高斯 [4, 6, 67]，DUSt3R、MASt3R 和 VGGT 则通过 Transformer 恢复几何与对应关系 [25, 55, 58]。Fast3R 和 CUT3R 探索图像集合的联合与递归处理 [56, 72]。TokenGS 通过可学习 token 明确解耦像素与高斯预测 [44]。本文的区别在于变量组织与推断任务：先通过条件生成补全背景和规范资产潜码，再将其解码为可复用场景。

驾驶场景推断。 STORM 预测动态高斯，并通过场景流聚合 [71]。DrivingForward 研究灵活的环视输入 [52]，DGGT 结合免位姿重建、时间可见性和运动引导的插值 [5]。StreetForward、PointForward 和 Instant NuRec 分别研究时序处理、点对齐表示及面向仿真的重建 [8, 21, 77]。DriveCompose 利用恢复的几何与运动，在持久支撑上汇集证据，使缺失内容在状态变化将其暴露之前得到补全。

规范与可驱动资产。 SMPL 和 SMPL-X 定义关节化人体模型 [29, 37]；Humans in 4D 和 WHAM 从图像恢复人体形状与运动 [15, 48]。HumanNeRF 和 Vid2Avatar 从视频重建人体 [16, 61]，GaussianAvatar、GauHuman、面向人体的 HUGS 和 3DGS-Avatar 使用可驱动高斯 [19, 20, 24, 41]。CanonicalFusion、FRESA、LHM 和 IDOL 从一张或多张图像推断可驱动人体资产 [42, 47, 57, 84]。UrbanCAD 则面向可控车辆资产 [30]。分割模型与合成人体数据提供互补的监督资源 [2, 23]。我们通过类别专属支撑，在整个补全过程中保留各资产的坐标系与运动绑定。

生成式重建与补全。 DreamFusion 将二维扩散先验蒸馏为三维内容 [39]；Zero-1-to-3、MVDream 和 CAT3D 为三维创建生成新视角或多视图图像 [12, 28, 46]。Objaverse 为三维学习提供显式资产数据 [9]。GenAssets 学习驾驶资产先验 [75]，结构化潜模型则联合编码几何与外观以支持生成 [65, 66]。CompleteSplat 补全高斯场景，JRM 通过潜空间流匹配融合未对齐的对象观测，LatentWorld 围绕实体组织语义占据生成 [26, 36, 64]。DriveCompose 在共享查询布局上补全可供 RGB 渲染的背景与规范资产，以已见潜码和自由空间约束未知内容。

可控驾驶视频生成。 MagicDrive、MagicDrive-V2、DriveDreamer 和 Vista 以场景结构或控制信号为条件生成驾驶图像 [11, 13, 14, 59]。GEN3C 使用三维信息进行相机可控的视频生成 [45]。StreetCrafter 和 HorizonForge 将可控扩散与场景重建或编辑结合 [60, 69]；WorldSplat 生成高斯并细化其渲染视频 [83]，DIFIX3D+ 则通过扩散图像细化改善重建 [62]。LPIPS 等感知相似性度量评估渲染外观 [79]。本文的建模还为补全内容指定持久身份、坐标系和运动规则，使后续状态变化复用同一份三维补全。

3. 方法

DriveCompose 按照后续控制的单位重建驾驶场景。我们首先恢复视觉证据，并在背景与规范资产支撑上聚合（第 3.1–3.2 节）；类型专属编解码器定义潜变量及其高斯解码（第 3.3 节）。条件流在该表示中补全缺失内容（第 3.4 节），随后装配场景并在观测视角下检查（第 3.5 节）。第 3.6 节说明各模块的学习过程。图 1 展示完整的构建与复用链路。

3.1. 问题定义与可组合表示

输入是同步图像 $\mathbf{I}_{t,v}$、时间戳、内参 $\mathbf{K}_v$ 及相机到传感器支架的标定 $\mathbf{T}^r_{c_v}$。支架位姿 $\mathbf{T}^w_r(t)$ 由日志提供，或以第一帧为参考估计；相机位姿为 $\mathbf{T}^w_{c_v}(t) = \mathbf{T}^w_r(t)\mathbf{T}^r_{c_v}$。变换作用于列向量，距离单位为米。图像采用校正后的针孔模型与线性 RGB；畸变、滚动快门和曝光差异在证据聚合之前处理。

输出为可组合高斯场景

$$\mathcal{W} = \big(\mathcal{G}_b, C_{\text{sky}}, V_{\text{sky}}, \{\mathcal{A}_a\}, \mathcal{B}, \xi\big), \tag{1}$$

其中，$\mathcal{G}_b$ 是世界空间背景，C_{sky} 是天空立方体贴图，V_{sky} 表示其有效方向，$\mathcal{A}_a$ 是身份 a 对应的规范资产，$\mathcal{B}$ 是有限空间支撑，ξ 是采样种子。每个高斯存储中心 $\boldsymbol{\mu}$、协方差 $\boldsymbol{\Sigma}$、不透明度 α 和方向外观，并通过高斯泼溅渲染 [22]。车辆保留局部几何与固定尺寸，人体另存静止骨架和蒙皮权重。场景状态控制这些固定组件的放置与关节运动。

我们通过条件生成建模缺失内容。令 $\mathcal{Q}$ 为恢复得到的查询布局，C 为其观测证据。补全模型采样未知潜码 $\mathbf{z}_u$，编解码器 D_ψ 将组合后的潜码解码为场景内容：

$$\mathbf{z}_u \sim p_\theta(\mathbf{z}_u \mid C, \mathcal{Q}), \qquad \mathcal{W} = D_\psi(\mathbf{z}^{\text{obs}}, \mathbf{z}_u; \mathcal{Q}). \tag{2}$$

解码器同时保留式 1 中固定的支撑、身份、绑定和天空有效域元数据。布局由输入日志建立，生成过程填充其中的未知内容。仿真时间改变解码场景的状态，流时间则描述潜码的构建过程。

3.2. 视觉恢复与规范聚合

共享视觉编码。 Driving Log Encoder 使用图像 patch 划分 [10] 与 Transformer 注意力 [54]，并加入图像位置、相机标定和时间编码。图像内部注意力提取空间特征，第二个 Transformer 在不同视角与时刻的池化图像摘要之间交互，再将上下文送回 patch 特征。几何头预测相机光轴深度、法线、置信度、语义和实例归属。置信度决定哪些预测可以提供表面证据。

可学习的资产查询读取全部图像 token，形成轨迹级身份槽；每个槽再读取单个时刻的特征以估计状态。由此，持久身份、尺寸和人体体型与逐帧变化的根位姿、关节旋转分开表示。根位姿先在支架坐标中预测，再通过 $\mathbf{T}^w_r(t)$ 映射到世界坐标。人体体型码选择对应身份的模板，模板包含固定规范支撑、骨架拓扑和绑定。身份槽容量限定可表示的资产数量。

查询支撑。 每个查询存储规范位置 $\mathbf{q}_i$、单元尺度 $\boldsymbol{\delta}_i$、类型 k_i、局部注意力组 g_i 和参考世界坐标 $\mathbf{q}^w_i$。可靠深度样本提供表面锚点。背景锚点与有限世界网格结合；车辆锚点经运动规范化后，与局部包围体中的网格结合。人体查询位于规范模板表面，天空查询为按立方体贴图面分组的单位方向。这些支撑为已见与缺失内容提供明确位置。

补全过程中布局保持固定。我们将查询放到第一帧的配置中以得到 $\mathbf{q}^w_i$，其中人体包含蒙皮变换。规范坐标标识组件内部的内容，参考世界坐标则使生成器能够联系该组件与周围场景。

运动感知证据聚合。 对于每份观测，我们将规范查询放到对应的资产状态：

$$\mathbf{p}_i(t) = \begin{cases} \mathbf{q}_i, & k_i = \text{background}, \\ \text{xyz}(\mathbf{T}^w_a(t)\bar{\mathbf{q}}_i), & k_i = \text{vehicle}, \\ \text{xyz}(\mathbf{T}^w_h(t)\mathbf{A}_i(t)\bar{\mathbf{q}}_i), & k_i = \text{human}. \end{cases} \tag{3}$$

其中，$\bar{\mathbf{q}}_i$ 为齐次坐标，xyz 提取欧氏坐标，$\mathbf{A}_i(t)$ 为加权蒙皮变换。投影到相机后，得到深度 $z_{i,t,v}$ 以及采样特征、深度和置信度 $(\boldsymbol{\phi}_{i,t,v}, d_{i,t,v}, c_{i,t,v})$。给定容差 $\eta = \eta_{\text{abs}} + \eta_{\text{rel}} d$，定义 $r = (d - z)/\eta$。表面观测需位于图像内、深度有效、置信度足够，与查询的实例身份一致，并满足 $|r| \le 1$。背景查询还要求静态背景语义。令 $a_{i,t,v}$ 表示这一有效性判断，则聚合为

$$w_{i,t,v} = a_{i,t,v} c_{i,t,v} \exp(-r^2_{i,t,v}/2), \tag{4}$$

$$\mathbf{h}_i = \frac{\sum_{t,v} w_{i,t,v} [\boldsymbol{\phi}_{i,t,v}; r_{i,t,v}]}{\max(\varepsilon, \sum_{t,v} w_{i,t,v})}. \tag{5}$$

查询置信度取被接受观测中的最大值。该筛选排除深度或实例身份与查询不一致的观测。仅当整个支撑单元都位于观测表面之前时，可靠射线才提供自由空间票：

$$z_{i,t,v} + \tfrac{1}{2}\|\boldsymbol{\delta}_i\|_2 < d_{i,t,v} - \eta. \tag{6}$$

有效观测权重之和大于零时，查询被标为已见。否则，足够的自由空间票将其标为自由空间，其余查询为未知。掩码 m_i、f_i 和 u_i 互斥，满足 $m_i + f_i + u_i = 1$。天空投影仅使用相机旋转，采用可靠天空语义和零深度残差，不接收自由空间票。

3.3. 结构化潜表示与解码

类型专属编码器将 $[\mathbf{h}_i; \mathbf{q}_i; \boldsymbol{\delta}_i]$ 映射为潜码 $\mathbf{z}_i \in \mathbb{R}^D$。各潜码维度相同，但查询仍保留类型、归属和坐标系。可见性掩码与置信度单独提供给补全模型，使部分观测与完整观测使用同一潜码编码映射。作为条件输入之前，未观测潜码置零；已见子集记为 $\mathbf{z}^{\text{obs}}$。

背景、车辆和人体解码器分别将每个查询独立映射为 K 个子高斯。查询 i 的子高斯 k 的中心被限制在其支撑单元内：

$$\boldsymbol{\mu}_{i,k} = \mathbf{q}_i + \tfrac{1}{2}\boldsymbol{\delta}_i \odot \tanh(\mathbf{b}_{i,k}). \tag{7}$$

正尺度与归一化旋转确定正定协方差。不透明度乘以 $1 - f_i$，使自由空间查询不能产生可见高斯。人体子高斯继承支撑查询的蒙皮权重；车辆子高斯共用所属资产的刚体变换。独立天空头预测立方体贴图颜色，并在 V_{sky} 中保留观测支持的方向域。

外观为每个颜色通道存储常数 RGB 项和三个局部方向系数，按 $c(d) = \max(0, c_0 + C_{\text{app}}\hat{d}_{\text{local}})$ 求值。解码逐查询执行：编解码器冻结后，不变的已见潜码产生不变的高斯参数。补全网络可以在查询之间交换上下文，同时保留这一解码性质。

3.4. 条件生成式重建

场景条件 Transformer。 我们采用 DiT 风格的 Transformer [38] 参数化补全速度场，并通过流匹配训练 [27]。输入结合当前潜码、已见潜码、掩码、置信度、类型、两种坐标描述、单元尺度和池化日志特征。背景按空间邻域分组，每个资产构成一个局部组，天空方向按立方体贴图面分组。局部自注意力分别处理各组，共享记忆随后交换上下文：

$$\mathbf{M}' = \mathbf{M} + \text{Attn}_r(\mathbf{M}, \widetilde{\mathbf{X}}, \widetilde{\mathbf{X}}), \tag{8}$$

$$\mathbf{U} = \widetilde{\mathbf{X}} + \text{Attn}_w(\text{LN}(\widetilde{\mathbf{X}}), \mathbf{M}', \mathbf{M}'). \tag{9}$$

其中，$\widetilde{\mathbf{X}}$ 拼接局部更新后的 token，注意力的三个参数依次为查询、键和值。$\mathbf{U}$ 后接残差前馈更新。流时间调制进入局部归一化，记忆由可学习 token 和日志摘要初始化。这一结构保留各类支撑的局部组织，同时使车辆或人体查询能够使用周围场景的信息。

流训练与约束采样。 令 $\mathbf{z}_1$ 为同一查询布局上的完整目标，$\boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I})$ 为高斯噪声。在流时间 $\tau \in [0, 1]$，我们仅对未知内容插值：

$$\mathbf{z}_\tau = m \odot \mathbf{z}^{\text{obs}} + u \odot [(1 - \tau)\boldsymbol{\epsilon} + \tau \mathbf{z}_1]. \tag{10}$$

已见潜码固定，自由空间潜码保持为零。未知查询的目标速度为 $\mathbf{z}_1 - \boldsymbol{\epsilon}$。引入可选损失选择器 s_i，目标为

$$\mathcal{L}_{\text{flow}} = \frac{\sum_i u_i s_i \|\mathbf{v}_\theta(\mathbf{z}_\tau, \tau, C)_i - (\mathbf{z}_{1,i} - \boldsymbol{\epsilon}_i)\|_2^2}{D \max(1, \sum_i u_i s_i)}. \tag{11}$$

式 10 中所有参与插值的未知查询都需要有效目标潜码，即使 s_i 将其中一部分排除在损失之外。该要求决定了第 3.6 节中完整目标的构造方式。

推理时，以噪声初始化未知潜码，并执行 S 次 Euler 步，步长为 $\Delta\tau = 1/S$：

$$\mathbf{z}^{j+1} = \Pi_{m,f}\left[\mathbf{z}^j + \Delta\tau\, u \odot \mathbf{v}_\theta(\mathbf{z}^j, j/S, C)\right]. \tag{12}$$

投影 $\Pi_{m,f}$ 在每步后恢复已见潜码，并清零自由空间潜码。种子确定一次采样补全。采样完成后再解码；相机或资产状态改变时复用得到的场景。

3.5. 场景装配与观测一致性

状态驱动渲染。 在线步 k 的状态 s_k 指定自车位姿、活动资产、车辆变换，以及人体根位姿与关节姿态。车辆的高斯中心和协方差变换为

$$\boldsymbol{\mu}^w = \mathbf{R}\boldsymbol{\mu}^a + \mathbf{t}, \qquad \boldsymbol{\Sigma}^w = \mathbf{R}\boldsymbol{\Sigma}^a\mathbf{R}^\top. \tag{13}$$

人体使用加权蒙皮变换 $\mathbf{A}_i = \sum_j w_{ij}\mathbf{P}_j\mathbf{B}_j^{-1}$，其中 $\mathbf{B}_j$ 和 $\mathbf{P}_j$ 是根坐标系中的静止与当前全局关节矩阵。变换后的中心为 $\bar{\boldsymbol{\mu}}_i^w = \mathbf{T}_h^w\mathbf{A}_i\bar{\boldsymbol{\mu}}_i^h$。组合局部线性映射 $\mathbf{J}_i$ 将协方差变换为 $\mathbf{J}_i\boldsymbol{\Sigma}_i\mathbf{J}_i^\top$，其逆 $\mathbf{J}_i^{-1}$ 将观察方向映回规范坐标，并在外观求值前归一化。接近奇异的蒙皮变换被拒绝。根位姿控制整体放置，根关节保持静止变换。

渲染器合成背景、活动资产和有效天空。外部策略与动力学可推进状态：

$$o_k = \mathcal{R}(\text{Compose}(\mathcal{W}, s_k), \mathcal{C}_k), \tag{14}$$

$$s_{k+1} = F_{\text{dyn}}(s_k, \pi(o_k), \Delta t). \tag{15}$$

重置或分支轨迹改变状态，同时保留 $\mathcal{W}$ 及其种子。有限空间支撑和有效天空方向决定覆盖范围。道路网格与碰撞代理采用和视觉资产一致的坐标，由物理后端维护。

构建场景的检查。 固定已见潜码能够保持解码参数，但新增表面仍可能遮挡这些内容。接受场景之前，我们在原始相机和资产状态下渲染补全候选及仅由已见查询构成的观测参考；参考渲染关闭全部非已见查询。令 $e_{\text{cand}}(p)$ 和 $e_{\text{obs}}(p)$ 为像素 p 处两者的平均绝对 RGB 误差，局部退化定义为

$$\Delta e(p) = \max\{e_{\text{cand}}(p) - e_{\text{obs}}(p), 0\}. \tag{16}$$

在可靠的非自车像素上平均这一量，避免某一区域的改善抵消其他区域的退化。接受条件还限制绝对 RGB 误差、有效深度误差和前景侵入率，并要求最低覆盖率；这些条件同时检查聚合结果和最差视图。前景侵入指新增渲染表面位于预测参考深度之前，且未出现在观测参考渲染中；其比例按可靠的非天空几何像素归一化。预测深度提供观测一致性参照，独立几何评测需要另一份参考。被拒绝的候选不导出复用。

3.6. 训练过程

学习分为四个阶段，使补全目标具有稳定含义。恢复模块确定证据与支撑，编解码器定义潜空间坐标，补全网络学习缺失潜码的分布。最后一个阶段将该分布适配到真实观测，同时保持其预测的表示不变。

阶段一：视觉恢复。 我们利用几何、语义、实例归属和资产标注监督图像编码器、几何头、身份槽及状态头。在整条轨迹上执行一次匹配，将槽分配给目标身份，并在所有时刻复用这一匹配。根平移、旋转和固定尺寸约束车辆状态；人体还需要身份一致的体型、模板绑定和关节监督。置信度根据几何可靠性训练或校准，依据包括深度、遮挡和重投影信息。这些目标建立聚合所需的坐标与身份一致性。

阶段二：潜表示拟合。 冻结恢复模块，训练类型专属编码器和解码器。每个样本先由部分输入建立一次查询布局，再将更完整的观测聚合到完全相同的位置、类型、顺序和归属上。这种对应关系是潜码补全的前提；若为完整场景另建布局，就会改变待预测变量。RGB、深度和轮廓监督训练解码场景，并结合空域不透明度与高斯尺度正则。高斯集合目标也可以采用集合匹配；仅在子高斯对应已知时使用直接坐标监督。

对于渲染图像 $\widehat{\mathbf{I}}$ 和深度 $\widehat{d}$，表示拟合及后续适配使用的观测项为带掩码的 L1 误差：

$$\mathcal{L}_{\text{rgb}} = \langle \|\widehat{\mathbf{I}} - \mathbf{I}\|_1/3 \rangle_{\Omega_I}, \tag{17}$$

$$\mathcal{L}_{\text{depth}} = \langle |\widehat{d} - d| \rangle_{\Omega_d}. \tag{18}$$

符号 $\langle \ell \rangle_\Omega$ 表示在掩码 Ω 的有效元素上取平均。深度目标 d 及其有效掩码由可用监督提供；它们不使观测检查成为独立的精度测量。

阶段三：补全先验学习。 冻结恢复模块和整个编解码器，将完整目标编码为 $\mathbf{z}_1$，并以式 11 优化补全 Transformer。完整场景或资产样本必须为式 10 中的每个未知查询提供有效目标，单独的部分真实日志不满足这一条件。完整场景提供跨组件上下文，完整资产提供对象内部内容；任一未知查询缺少有效目标潜码时，有效性检查会拒绝该样本。

条件掩码模拟连续未见表面、背景空洞、长时间遮挡、短轨迹和不均匀相机覆盖。对恢复位姿、实例归属和置信度施加扰动，使生成器接触不完美证据。该阶段中，目标潜码和输入潜码始终使用相同的编解码器与查询定义。

阶段四：通过渲染适配。 保持恢复模块与编解码器冻结，通过流积分、解码和渲染反向传播，仅更新补全网络。部分真实日志监督渲染观测，已知自由射线和几何后端样本约束新增内容：

$$\mathcal{L}_{\text{scene}} = \lambda_e \langle a_r^2 \rangle + \lambda_g \langle [-d_f]_+^2 \rangle + \lambda_c \langle |d_f| \rangle_{\Omega_c} + \lambda_p \langle [d_p]_+^2 \rangle, \tag{19}$$

$$\mathcal{L}_{\text{adapt}} = \lambda_r \mathcal{L}_{\text{rgb}} + \lambda_d \mathcal{L}_{\text{depth}} + \mathcal{L}_{\text{scene}}. \tag{20}$$

其中，a_r 为已知自由射线上的累积不透明度，d_f 为脚点有符号距离且地面上方为正，Ω_c 选择接地脚点，d_p 为穿透深度且另一资产内部为正。采用 $[x]_+ = \max(x, 0)$；无掩码的平均在提供的样本上计算。系数 λ 为有效约束赋权。自由射线不透明度在查询级空域掩码之外，进一步约束新增内容的渲染影响。

完整目标样本可与真实日志适配交替使用以保留先验，得到 $\mathcal{L}_4 = \lambda_f \mathcal{L}_{\text{flow}} + \mathcal{L}_{\text{adapt}}$，各项仅作用于具备所需目标的样本。已见锚点由冻结解码和潜码硬恢复维持。

推理。 日志只编码一次，随后固定支撑与证据，由补全模型积分未知潜码。解码并检查场景后，导出固定内容；运行时渲染仅使用这些内容和指定状态。恢复模块、编解码器、补全网络与查询定义的版本一并记录，因为前两者的变化会改变补全网络学习的潜分布。

4. 实验

5. 局限

DriveCompose 表示固定照明下的有限区域。恢复深度、身份或规范姿态中的错误可能进入已见锚点，并对后续补全施加错误约束。车辆采用刚体模型，人体保留固定衣物体积和骨架绑定；超出模板支撑范围的姿态可能暴露缺失几何。参考世界坐标提供构建时的上下文，编辑后的接触与碰撞有效性则依赖物理后端。观测检查衡量与已记录证据的一致性，证据之外的几何和照明变化下的外观需要额外监督或建模。

6. 结论

本文提出 DriveCompose，用于可组合驾驶场景的生成式重建。结构化潜表示将图像证据汇入背景与规范资产，使缺失内容在生成之前就具有明确的位置、身份和运动模型。条件补全利用场景上下文并保留已观测证据，类型解码输出可复用高斯场景的组成部分。内容构建与状态驱动装配相分离，使后续相机和资产状态变化能够使用同一次补全结果。

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